LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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Not every countable group is MF in the operator-norm approximation sense. We prove a one-sided compression criterion that places normal property (T) subgroups in the MF radical, and normal generation can make that radical equal the whole group. An elementary group over the binary Leavitt algebra satisfies the criterion, giving a simple property (T) group that is not MF. We also prove exact pullback formulas showing that an MF-invisible kernel preserves the complete family of MF quotients above it.

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1. Introduction

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum. Equivalently, an MF group admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements: for each $g \neq 1$, the quantity $\limsup_n \|V_n(g) - 1\|$ is positive, with a lower bound that may depend on g. Throughout, *MF group* means the operator-norm notion introduced by Carrión–Dadarlat–Eckhardt [CDE, Definition 2.7].

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

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Proposition 1.1 below proves that G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF-kernel closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap\{\ker f : N \leq \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 1.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism of G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R as $(x_j)_{j\geq 1}$, and enumerate the pairs in $(G/R)^2$. For each x_j, the definition of R supplies a corona homomorphism whose value at x_j has positive distance from the identity. Every such homomorphism kills R and therefore descends to G/R. Choose coordinate unitary lifts, using polar correction as in Lemma 4.1. At stage n, take a direct sum of one coordinate from each of the first n models. Choose each coordinate far enough out that the first n multiplication defects are at most $1/n$ and the designated value at x_j retains at least half of its corona-norm separation. Such coordinates exist arbitrarily far out: the defects converge to zero, while the quotient norm is a limsup. The resulting operator-norm asymptotic representation detects every x_j, so its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all MF-target maps killing N is N, which is the asserted closure criterion. □

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF; if G has property (T) *and* $\mathfrak{D}_G(L) = G$, *then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

Corollary 3.4 puts $\mathfrak{D}_G(L)$ in the kernel detected by normalized Hilbert–Schmidt norm. If a corona representation were nontrivial on K, normality of K would make its Kazhdan projection commute with the image of G. The complementary corner would then contain no K-fixed vectors. A Kazhdan inequality gives an element of K whose Hilbert–Schmidt distance from the identity is bounded below there, contrary to the first conclusion. Independently, Theorem 2.1 shows that every finite-dimensional linear representation over every field kills $\mathfrak{D}_G(L)$.

Whether every countable group is MF was asked in the group-MF literature and remained open [CDE, Kor, Th, Sh]. Applying Theorem A to the binary Leavitt construction below answers the question.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put* $R = L_{\mathbb{F}_2}(1,2)$ *and* $H = \mathrm{EL}_{12}(R)$. *Then H is nontrivial, simple, has property* (T), *and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \mathrm{EL}_3(R)$. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the elementary commutator relations show directly that d normally generates H. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator-norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator-norm control used to construct the conjugation representation in Theorem 3.3.

Full MF radicals determine prescribed MF-visible quotients as follows.

Theorem C (prescribed MF-visible quotients)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q \colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. Carrión–Dadarlat–Eckhardt introduced the group-MF framework used here [CDE]; Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves MF permanence for doubles $G *_H G$ of MF groups and, separately, MF full group C^*-algebra results for several classes of amalgamated and semidirect products [Sh]. The operator-norm case remained open after non-approximability had been established for finite Schatten norms [DGLT, LO, Th]. Bachner–Dogon–Lubotzky study an operator–Hilbert–Schmidt stability condition as a possible source of non-MF groups [BDL]. Here operator-norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. Its Kazhdan projection makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Section 4 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's work gives general normal-subgroup theorems for linear groups over exchange rings [Pre]; the simplicity proof below instead extracts an elementary root directly from a nonidentity normal element. Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and

R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. □

The finite-dimensional hypothesis is essential here. On an infinite-dimensional space an injective endomorphism of the commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity at the level of fixed-space projections.

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator-norm asymptotic representations of G. An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty\to 2}(G)$.

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 3.1. *For every sequence (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^* x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^* x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \leq q$ satisfy $p = q$: if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1-q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. □

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\mathrm{Fix}\,\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\,\mathrm{Fix}\,\pi(L) \subseteq \mathrm{Fix}\,\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator-norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V,L)) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V,L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

so the operator-norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\mathrm{Fix}\,\sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \leq L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge ordinarily to zero. Thus $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V,L)$, which proves the equality. □

Corollary 3.4. *Let $L \leq G$ have property* (T)*, let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V,L)$. By Theorem 3.3, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\left\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\right\| \longrightarrow 0.$$

Therefore

$$\left\|V_n([ucu^{-1},\ell]) - 1\right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. Normal Kazhdan subgroups and the MF radical

Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona representation of G. Normality of K implies that p commutes with the image of G. Hence $q = 1 - p$ also commutes with that image, and the corner $q\mathcal{Q}_{\mathbf{d}}q$ has no nonzero K-fixed vectors.

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator-norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$. Projection lifting follows by spectral rounding. Unitary lifting follows by polar-correcting any lift, since its two unitarity defects converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator-norm asymptotic representation. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates. □

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable, let $D \le R_{\infty\to 2}(G)$, and let $K \trianglelefteq G$ have property* (T) *with $K \le D$. Then*

$$K \le \mathrm{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator-norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Thus p is the projection onto the K-fixed vectors in every faithful representation of the corona. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\,\mathrm{Fix}\,\Theta(K)$ and $\mathrm{Fix}\,\Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero, since $p = 1$ would imply that every element of K has trivial image. Lemma 4.1 gives an operator-norm asymptotic representation (W_n) on nonzero corners $q_n M_{d_n}(\mathbb{C}) q_n$.

Choose a finite symmetric Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. In the corner $q\mathcal{Q}_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (q\Theta(s)q - q)^* (q\Theta(s)q - q)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|} q.$$

Indeed, in every representation of the corner there are no K-fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - q_n)^* (W_n(s) - q_n)$$

represent b. Taking the normalized trace on $q_n M_{d_n}(\mathbb{C}) q_n$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - q_n\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)q_n$ converges to zero in operator norm, which gives the displayed trace inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0 \in K \leq D \leq R_{\infty\to 2}(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 3.4 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$

Theorem 4.2 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. The two stated consequences follow by taking $K \neq 1$, or $K = G = \mathfrak{D}_G(L)$, respectively.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. □

Proposition 4.3 (functoriality and normal generation)**.** *Let $L \leq G$ and put $D = \mathfrak{D}_G(L)$.*

If $f\colon G \to Q$ is a homomorphism, then

$$f(D) \leq \mathfrak{D}_{f(G)}(f(L)). \tag{4}$$

If $S \leq G$ is a nontrivial simple subgroup and $D \cap S \neq 1$, then $S \leq D$. If in addition the normal closure of S in G is all of G, then $D = G$. If moreover G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = G.$$

Suppose instead that f is onto, $S \leq D$ is simple, $f(S) \neq 1$, and $f(S)$ normally generates Q. Then

$$\mathfrak{D}_Q(f(L)) = Q.$$

If, in addition, G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(Q) = Q.$$

Proof. If $u \in \mathrm{Comp}_G(L)$, then $f(u)f(L)f(u)^{-1} \leq f(L)$; if $c \in C_G(L)$, then $f(c) \in C_{f(G)}(f(L))$. Therefore f takes every defining generator of D into a defining generator of $\mathfrak{D}_{f(G)}(f(L))$. This proves (4).

The subgroup D is normal in G, so $D \cap S$ is normal in S. Simplicity and $D \cap S \neq 1$ give $D \cap S = S$, hence $S \leq D$. Since D is normal in G, it then contains the normal closure of S. Under the second hypothesis this is G. The direct radical assertion is Theorem A with $K = G$.

For the image statement, (4) gives $f(S) \leq \mathfrak{D}_Q(f(L))$. The latter subgroup is normal in Q, so it contains the normal closure of $f(S)$ and is therefore all of Q. Property (T) passes to quotients, so the final hypotheses give property (T) for both $f(L)$ and Q. Applying Theorem A inside Q, with $K = Q$, proves the last assertion. $\square$

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{5}$$

In particular $q \neq 0$.

We use indices $0, \ldots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{6}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q s_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$ proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_3 to pI_3.

Set

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A direct calculation using (2) gives $XY = YX = I_6$. Put

$$\tau = \operatorname{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{7}$$

Writing $I = I_6$, the Whitehead factorization is

(8)

$$\operatorname{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

Each block-unipotent factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block may be inserted one entry at a time, and the corresponding matrix units have pairwise zero products. Since $Y = X^{-1}$, (8) proves $\tau \in \mathrm{EL}_{12}(R)$.

Let $H = \mathrm{EL}_{12}(R)$ and let $L = \mathrm{EL}_3(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ, Theorem 1.1]. Block multiplication yields

$$\tau \operatorname{diag}(A, I_9)\, \tau^{-1} = \operatorname{diag}(\Psi(A), I_9). \tag{9}$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$\tau L \tau^{-1} \leq L. \tag{10}$$

The field $\mathbb{F}_2$ is used in the diagonal branch of the simplicity argument below: the only central unit of R is then 1. The dimension 12 comes from applying the 6×6 Whitehead factorization to the 3×3 Kazhdan corner. We do not claim that 12 is minimal.

6. The binary Leavitt group has full MF radical

Proposition 6.1. *The group $H = \mathrm{EL}_{12}(R)$ is simple.*

Proof. We use two concrete ring facts. First, for every $0 \neq x \in R$ there are $a, b \in R$ such that

$$axb = 1. \tag{11}$$

This is the form of pure infiniteness needed here [AA]. Second, the only central unit of R is 1: indeed $Z(R) = \mathbb{F}_2$ by [AC, Corollary 4.3] and $\mathbb{F}_2^\times = \{1\}$.

We first record that a nonzero root normally generates the elementary group. More generally, let S be a unital ring, let $n \geq 3$, and suppose $a, x, b \in S$ satisfy $axb = 1$. If a normal subgroup of $\mathrm{EL}_n(S)$ contains $e_{ij}(x)$, choose k different from i and j. The elementary commutator relations give

$$[e_{ij}(x), e_{jk}(b)] = e_{ik}(xb), \qquad [e_{ji}(a), e_{ik}(xb)] = e_{jk}(1),$$

so the normal subgroup contains a unit root. From any $e_{uv}(1)$ in the normal subgroup and any third index w, the relations

$$[e_{wu}(r), e_{uv}(1)] = e_{wv}(r), \qquad [e_{uv}(1), e_{vw}(r)] = e_{uw}(r)$$

put the corresponding source and target moves, with arbitrary coefficient $r \in S$, in the normal subgroup. Repeating these moves, using a third index when needed, gives every elementary generator. Thus, by (11), it remains to prove that every nontrivial normal subgroup $N \trianglelefteq H$ contains a nonzero elementary root.

We also need the following coefficient-separation consequence of the Leavitt relations. If $r, s \in R$ are nonzero, then there are $a, b \in R$ such that

$$arb = 0, \qquad bsar \neq 0. \tag{12}$$

Here is a direct proof. For every $w \in R$ there are $x, y \in R$ with $xwy = 0$ and $yx \neq 0$. This is immediate with $x = y = 1$ if $w = 0$. Otherwise choose c, d with $cwd = 1$ and put $x = t_1 c$, $y = ds_0$. Then $xwy = t_1 s_0 = 0$. Moreover $s_0 t_1 \neq 0$, since

$$1 = (t_0 s_0)(t_1 s_1) = t_0(s_0 t_1)s_1,$$

and $yx = ds_0 t_1 c = 0$ would imply

$$s_0 t_1 = (cwd)(s_0 t_1)(cwd) = (cw)\big((ds_0)(t_1 c)\big)(wd) = 0.$$

To obtain (12), choose factorizations

$$c_r\, r\, d_r = 1, \quad e_d\, d_r\, f_d = 1, \quad e_s\, s\, f_s = 1, \quad e_t(c_r r)f_t = 1,$$

apply the preceding construction to $w = e_t f_d$, and set

$$a = f_s x e_t c_r, \qquad b = d_r f_d y e_s.$$

Then $arb = 0$. If $bsar = 0$, multiplication on the left by e_d and on the right by f_t gives $yx = 0$, a contradiction.

Now let $N \trianglelefteq H$ be nontrivial and choose $1 \neq g \in N$. Suppose first that g is diagonal. Its inverse is also diagonal. If g commuted with every $e_{ij}(a)$, then commuting with the $e_{ij}(1)$ would force $g = \lambda I_{12}$, and commuting with arbitrary $e_{ij}(a)$ would make λ a central unit. Hence $g = 1$, a contradiction. Thus for some $i \neq j$ and $a \in R$, conjugation by g sends $e_{ij}(a)$ to $e_{ij}(y)$ with $y \neq a$. Consequently

$$[g, e_{ij}(a)] = e_{ij}(y - a) \in N$$

is a nonzero root.

Suppose next that g is not diagonal, and choose $\ell \neq i$ with $g_{\ell i} \neq 0$. Put $r = (g^{-1})_{\ell i}$. If $r = 0$, set

$$A = gE_{i\ell}g^{-1}, \qquad B = E_{i\ell}.$$

Then $AB = 0$, and the double commutator

$$[ge_{i\ell}(1)g^{-1}, e_{i\ell}(1)] = 1 - BA \tag{13}$$

belongs to N. Its defect $-BA$ is square-zero and supported in row i. There is an m for which $g_{\ell i}(g^{-1})_{\ell m} \neq 0$: otherwise the (ℓ, ℓ) entry of $g^{-1}g = 1$ would give $g_{\ell i} = 0$. Thus the (i, m) entry of the defect in (13) is nonzero.

It remains to treat $r \neq 0$. Apply (12) to r and $s = g_{\ell i}$, obtaining a, b with $arb = 0$ and $bsar \neq 0$. This time put

$$A = g(aE_{i\ell})g^{-1}, \qquad B = bE_{i\ell}.$$

Again $AB = 0$, because its potentially nonzero coefficient contains arb. As the two elementary roots commute, reduction modulo N shows that

$$[ge_{i\ell}(a)g^{-1}, e_{i\ell}(b)] = 1 - BA \in N. \tag{14}$$

The defect is square-zero and supported in row i, while its (i, i) entry is $-bsar \neq 0$.

Both off-diagonal cases therefore produce an element $1 + v \in N$ with $v^2 = 0$, supported in a single row i, and with some $v_{im} \neq 0$. Choose n different from both i and m (with only $n \neq i$ needed when $m = i$). Direct multiplication gives the row-extraction identity

$$[1 + v, e_{mn}(1)] = e_{in}(v_{im}) \in N.$$

This is a nonzero root, so the normal-generation argument above gives $N = H$. Hence every normal subgroup of H is trivial or all of H. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. □

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c\tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\langle d\rangle\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 3×3 corner has both indices in $\{0, 1, 2\}$. The elementary commutator relations show that it commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (7) gives

$$\tau c\tau^{-1} = e_{01}(q)e_{34}(1). \tag{15}$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Since $t_1 q s_1 = 1$, the preceding sandwiched-root argument gives $\langle\langle d\rangle\rangle_H = H$ directly. □

Proof of Theorem B. The finitely generated $\mathbb{F}_2$-algebra R is countable, and hence so is H. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (10), the centrality of c relative to L, and Proposition 6.2 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 6.2 gives $\langle\langle d\rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Independently of this radical calculation, Proposition 6.1 shows that H is nontrivial and simple.

If M is MF and $f \colon H \to M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. □

7. MF-visible quotients

The MF radical is exact across every surjection whose kernel is already MF-invisible.

Proposition 7.1 (full-kernel pullback). *Let $f\colon G \to Q$ be a surjective homomorphism of countable groups. If $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

In particular, the conclusion holds whenever $\mathrm{Rad}_{\mathrm{MF}}(\ker f) = \ker f$. More generally, for every normal subgroup $N \trianglelefteq G$,

$$\mathrm{cl}^{G}_{\mathrm{MF}}(N) = f^{-1}\left(\mathrm{cl}^{Q}_{\mathrm{MF}}(f(N))\right).$$

Consequently,

$$G/N \text{ is MF} \iff \ker f \le N \text{ and } Q/f(N) \text{ is MF.}$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f. Conversely, composing a corona homomorphism of Q with f gives one of G. The two families of kernels have the displayed intersection. For the final assertion, functoriality of the MF radical under the inclusion $\ker f \to G$ gives $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N, gives the closure formula. If this closure equals N, then it contains $\ker f$, and mapping the equality onto Q shows that $\mathrm{cl}^{Q}_{\mathrm{MF}}(f(N)) = f(N)$. Conversely, these two conditions give

$$\mathrm{cl}^{G}_{\mathrm{MF}}(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 1.1, applied in G and Q. ☐

Thus image under f and inverse image under f give mutually inverse, inclusion-preserving correspondences between the normal subgroups that define MF quotients of G and of Q. The map induced by f on the largest MF-visible quotients is an isomorphism:

$$G/\mathrm{Rad}_{\mathrm{MF}}(G) \cong Q/\mathrm{Rad}_{\mathrm{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with MF-invisible kernels.

Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \ne d \in B$ normally generate B. Put $A = \langle d \rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{16}$$

where $A \to Q \times A$ is the inclusion into the second factor. The maps from B and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A. They therefore define an epimorphism

$$\pi_Q\colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (16) are injective; in particular, $d \neq 1$ in W_Q.

Proposition 7.2 (universal factorization). *Let T be a group such that every homomorphism $B \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\mathrm{Hom}(Q,T) \longrightarrow \mathrm{Hom}(W_Q,T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f\colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q,a) \in Q \times A$, we then have

$$f(q,a) = f(q,1)f(1,a) = f(q,1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills B, since $\langle\langle d \rangle\rangle_B = B$, and kills the second factor A in $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. □

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\mathrm{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.2 therefore gives the asserted bijection for every MF group M. In addition, the definition of $\mathrm{Rad}_{\mathrm{MF}}(B) = B$ says directly that every homomorphism from B to $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is trivial. Applying Proposition 7.2 with $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ shows that every corona homomorphism from W_Q factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

□

If $N \trianglelefteq W_Q$, the same factorization gives

$$\mathrm{cl}_{\mathrm{MF}}^{W_Q}(N) = \pi_Q^{-1}\left(\mathrm{cl}_{\mathrm{MF}}^{Q}(\pi_Q(N))\right). \tag{17}$$

Indeed, an MF-target homomorphism from W_Q kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (17) give

$$W_Q/N \text{ is MF} \quad \iff \quad \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \leq N$, then $N = \pi_Q^{-1}(\pi_Q(N))$; this is the only additional observation needed for the displayed equivalence.

Taking $Q = \mathbb{Z}$ makes the quotient criterion especially concrete: for every normal subgroup $N \trianglelefteq W_{\mathbb{Z}}$,

$$W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N. \tag{18}$$

Indeed, every quotient of $\mathbb{Z}$ is MF, while $\ker \pi_{\mathbb{Z}} = \langle\langle d \rangle\rangle_{W_{\mathbb{Z}}}$.

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References

[AA] G. Abrams and G. Aranda Pino, *Purely infinite simple Leavitt path algebras*, J. Pure Appl. Algebra **207** (2006), no. 3, 553–563, doi:10.1016/j.jpaa.2005.10.010.

[AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.

[AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, arXiv:2608.05362.

[AC] G. Aranda Pino and K. Crow, *The center of a Leavitt path algebra*, Rev. Mat. Iberoam. **27** (2011), no. 2, 621–644, doi:10.4171/RMI/649.

[Ara] P. Ara, *The exchange property for purely infinite simple rings*, Proc. Amer. Math. Soc. **132** (2004), no. 9, 2543–2547, doi:10.1090/S0002-9939-04-07369-1.

[BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's Property (T)*, Cambridge University Press, 2008.

[BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1-approximation of groups*, J. Algebra **702** (2026), 235–243, doi:10.1016/j.jalgebra.2026.04.041, arXiv:2508.17392.

[CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^*-algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, doi:10.1016/j.jfa.2013.04.004.

[Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K-theory*, Adv. Math. **384** (2021), Paper No. 107722, doi:10.1016/j.aim.2021.107722.

[DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, doi:10.1017/fms.2020.5.

[EJZ] M. Ershov and A. Jaikin-Zapirain, *Property* (T) *for noncommutative universal lattices*, *Invent. Math.* **179** (2010), 303–347, doi:10.1007/s00222-009-0218-2.

[Kor] A. I. Korchagin, *The MF-property for countable discrete groups*, Math. Notes **102** (2017), no. 2, 198–211, doi:10.1134/S0001434617070227, arXiv:1704.06906.

[Lev] W. G. Leavitt, *The module type of a ring*, *Trans. Amer. Math. Soc.* **103** (1962), 113–130, doi:10.1090/S0002-9947-1962-0132764-6.

[LO] A. Lubotzky and I. Oppenheim, *Non p-norm approximated groups*, J. Anal. Math. **141** (2020), no. 1, 305–321, doi:10.1007/s11854-020-0119-2.

[KT] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan's property* (T), preprint, 2019, arXiv:1901.03963.

[Pre] R. Preusser, *On general linear groups over exchange rings*, Linear and Multilinear Algebra **70** (2022), no. 4, 705–713, doi:10.1080/03081087.2020.1743636.

[Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, arXiv:2603.13564.

[Th] A. Thom, *Finitary approximations of groups and their applications*, Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018, arXiv:1712.01052.